Batterjee Medical College
For Science and Technology

BMC

Sleep Apnea

An approach to a Case of fatigue

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BATTERJEE MEDICAL COLLEGE

Learning Objectives

Knowledge:

- Define Obstructive Sleep Apnea (OSA), Central Sleep Apnea (CSA), and Obesity Hypoventilation Syndrome (OHS/Pickwickian Syndrome), and outline the stages of normal sleep.
- Detail the physiological mechanisms leading to airway collapse in OSA, brainstem respiratory center failure in CSA, and hypoventilation in OHS.
- Explain how chronic nocturnal hypoxia and hypercapnia trigger hypoxic pulmonary vasoconstriction, pulmonary hypertension, cor pulmonale (right-sided heart failure), secondary hypertension, and renal compensation.
- Describe key predisposing factors and clinical manifestations including loud snoring, daytime sleepiness, early morning headaches, witnessed apneas, and Cheyne-Stokes breathing patterns.

Skills:

- Utilize standardized screening tools such as the STOP-Bang questionnaire to stratify patient risk.
- Analyze diagnostic modalities, specifically overnight polysomnography parameters to differentiate between obstructive and central events.
- Select appropriate therapies based on patient diagnosis and severity, ranging from lifestyle modifications and Continuous Positive Airway Pressure (CPAP/PAP) to non-invasive positive pressure ventilation (NIPPV) or surgical options.
- Apply diagnostic reasoning to evaluate complex presentations of sleep-disordered breathing and its systemic sequelae.

Value:

- Demonstrate a professional attitude in adherence to training place rules, regulations, and institutional standards during clinical learning.
- Recognize the severe impact of chronic fatigue, daytime sleepiness, and sleep disorders on a patient's overall quality of life and safety.
- Value the approach required to manage sleep apnea, coordinating care across internal medicine, sleep specialty centers, surgical teams, and lifestyle counselors.

Main Subtopics

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- Obstructive Sleep Apnea
- Central Sleep Apnea
- Obesity Hypoventilation Syndrome

Sleep

WHAT IS SLEEP?



Sleep is a natural state of rest that we enter during the night.

During sleep, our body and mind relax and recover. It is a vital process for our health and well-being.

During sleep, the body performs various functions, such as tissue repair and regeneration, memory consolidation, and hormonal balance.

Additionally, sleep helps us replenish energy, improve our mood, and strengthen the immune system. Getting enough sleep and having quality sleep is essential to staying healthy and functioning properly in our daily lives.

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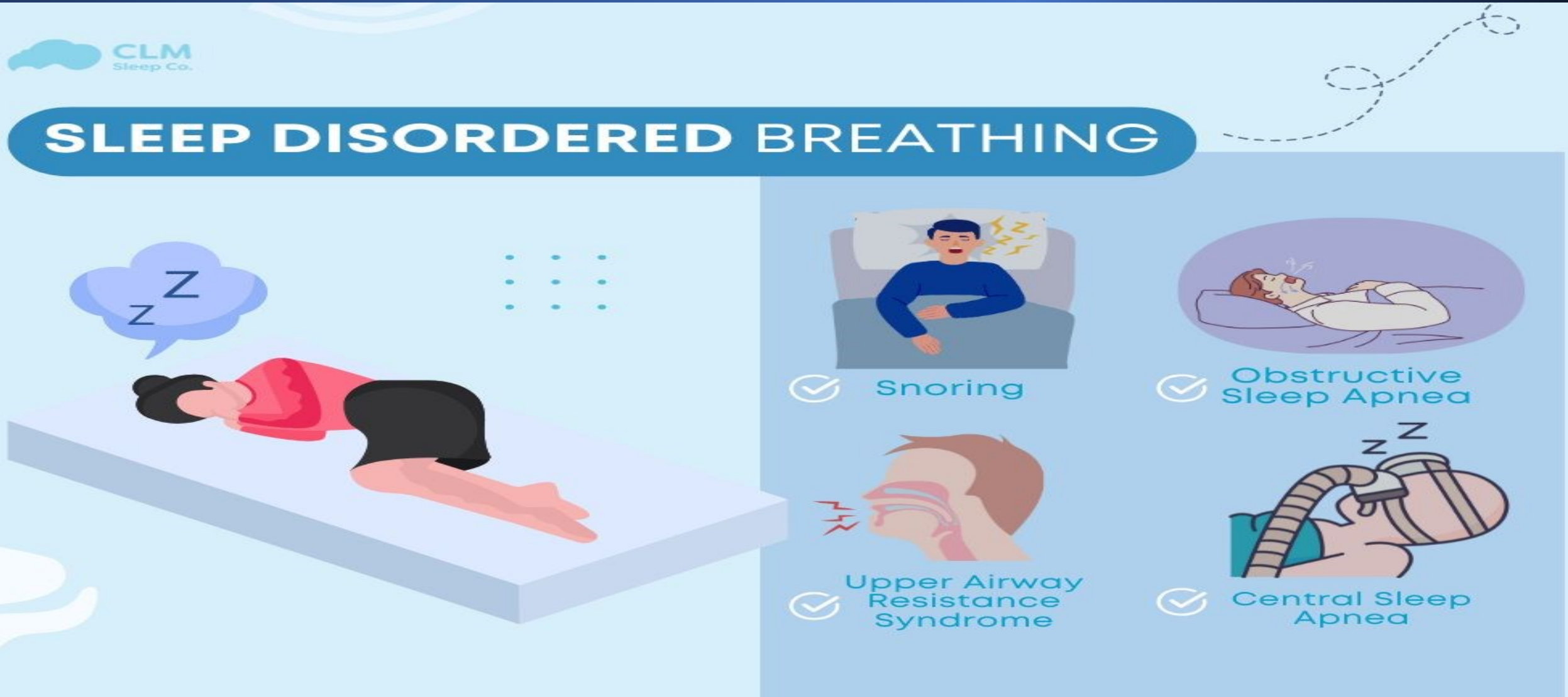
Sleep


Stage 1
In stage 1, we experience a light transitional sleep. This is where drowsiness and sleep begin.
Stage 2
In stage 2 more stable sleep occurs. Chemicals produced in the brain block the senses making it difficult to be woken.
Stage 3
Stage 3 is deep sleep. Growth horone is released during this stage. Most stage 3 sleep occurs in the first third of the night.
REM
REM sleep revitalizes the memory. In this stage, brain activity is very high and intense dreaming is likely to occur.

Sleep Disorders



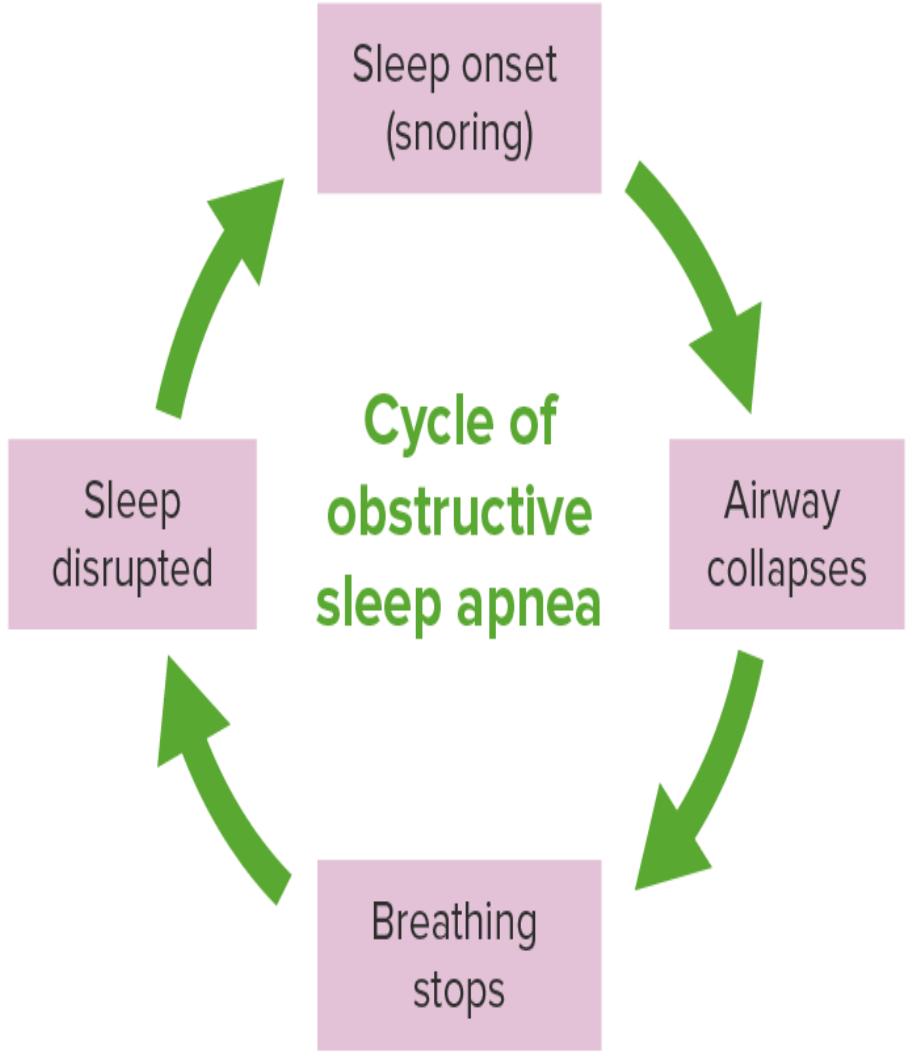
Sleep Disorders



Abnormal Breathing events in adults

Abnormal Breathing Events:

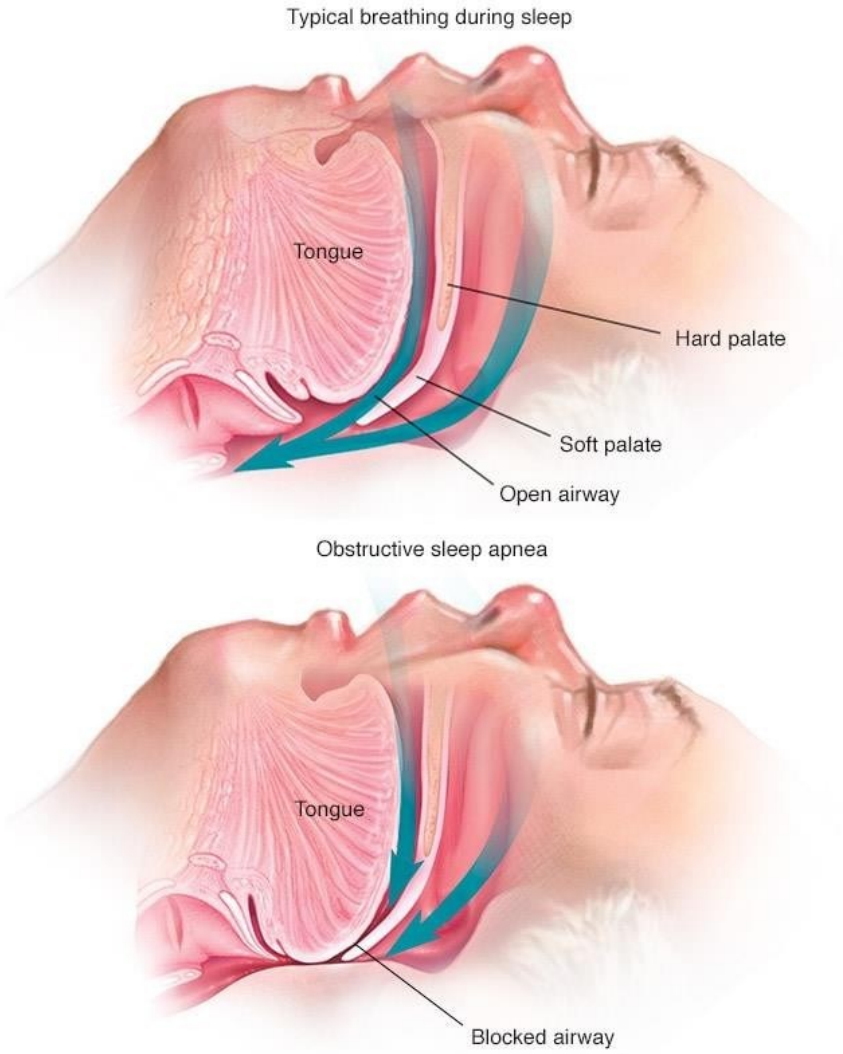
- **Apnea:** complete or nearly complete (≥ 90%) cessation of inspiratory airflow for ≥ 10 seconds
- **Hypopnea:** airflow reduction by ≥ 30% of the pre-event baseline for ≥ 10 seconds in combination with either desaturation by ≥ 3% or arousal from sleep
- **Respiratory effort-related arousal (RERA):** arousal from sleep due to increased respiratory effort or reduced airflow for ≥ 10 seconds without significant hypopnea or apnea



Obstructive Sleep Apnea (OSA)

Definition:

- Sleep-related breathing disorder in which airflow significantly decreases or ceases due to upper airway obstruction
- Intermittent obstruction of the air flow (typically at the level of the oropharynx)
- produces periods of apnea during sleep.
- Each apneic period is usually 20 to 30 seconds long (but may be longer)
- Apneic episodes results in hypoxia, which arouses the patient from sleep.
- This occurs multiple (sometimes hundreds of) times overnight.
- Most common sleep disorder



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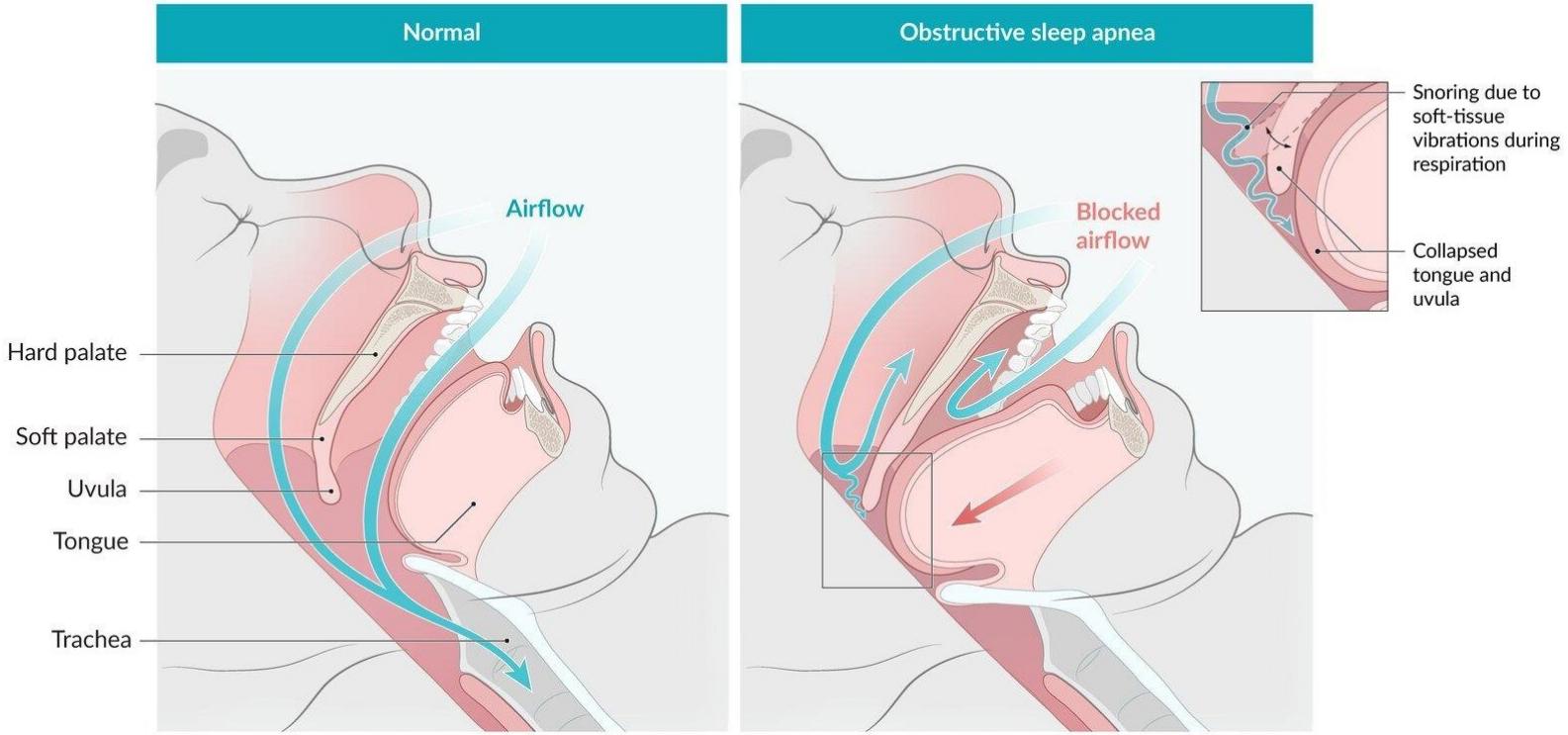
Obstructive Sleep Apnea (OSA)

Etiology:

- collapse of the pharyngeal muscles during sleep

Risk factors:

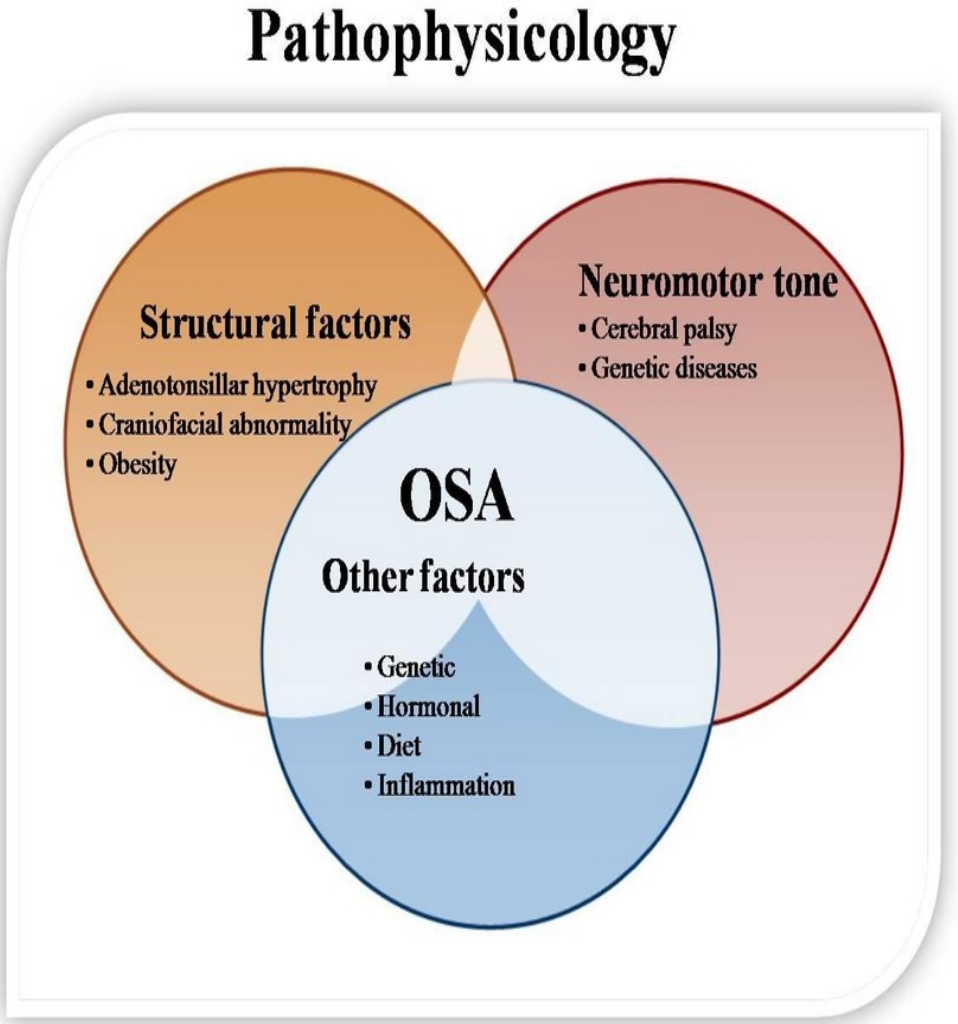
- Obesity
- Structural abnormalities
- Alcohol
- Smoking
- Acromegaly
- Hypothyroidism



Obstructive Sleep Apnea (OSA)

Pathophysiology

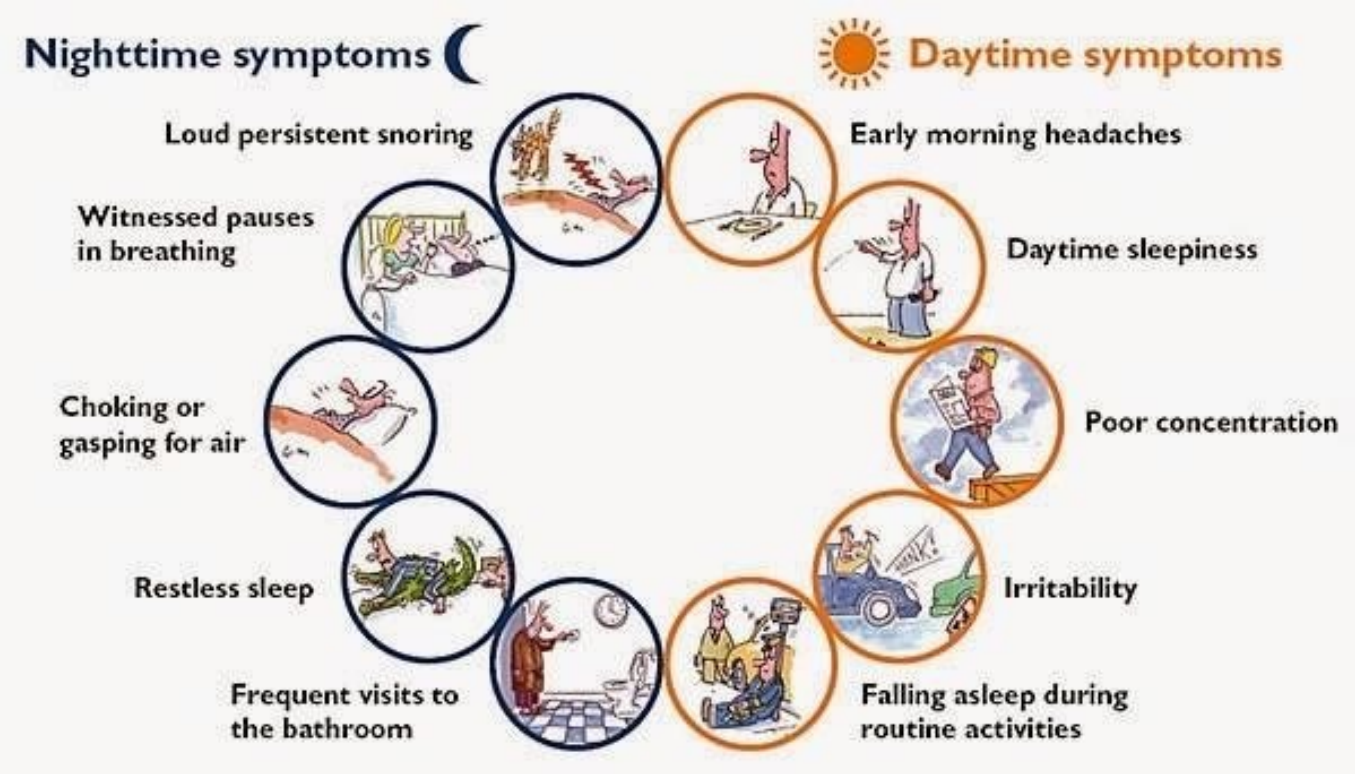
- Obstruction of the upper airway → apnea → ↓ PaO2 &↑ partial PaCO2 which leads to:
- ↑ Hypoxic pulmonary vasoconstriction → ↑ pulmonary hypertension → cor pulmonale
- ↑ Sympathetic activity → secondary hypertension
- Respiratory acidosis → renal compensation → increased HCO3 retention



Obstructive Sleep Apnea (OSA)

Clinical Features:

- Snoring
- Daytime sleepiness
- Personality changes,
- Decreased intellectual function
- Pulmonary HTN
- Cardiac arrhythmias
- Morning headaches
- Polycythemia
- Hypertension
- Decrease Libido



Obstructive Sleep Apnea (OSA)

Diagnostics:

- Sleep history, including third-party reports
- Evaluate for co morbidities
- Screening for OSA (standardized questionnaire)
- Polysomnography (Sleep studies) are required to confirm the diagnosis



Obstructive Sleep Apnea (OSA)

STOP-Bang questionnaire

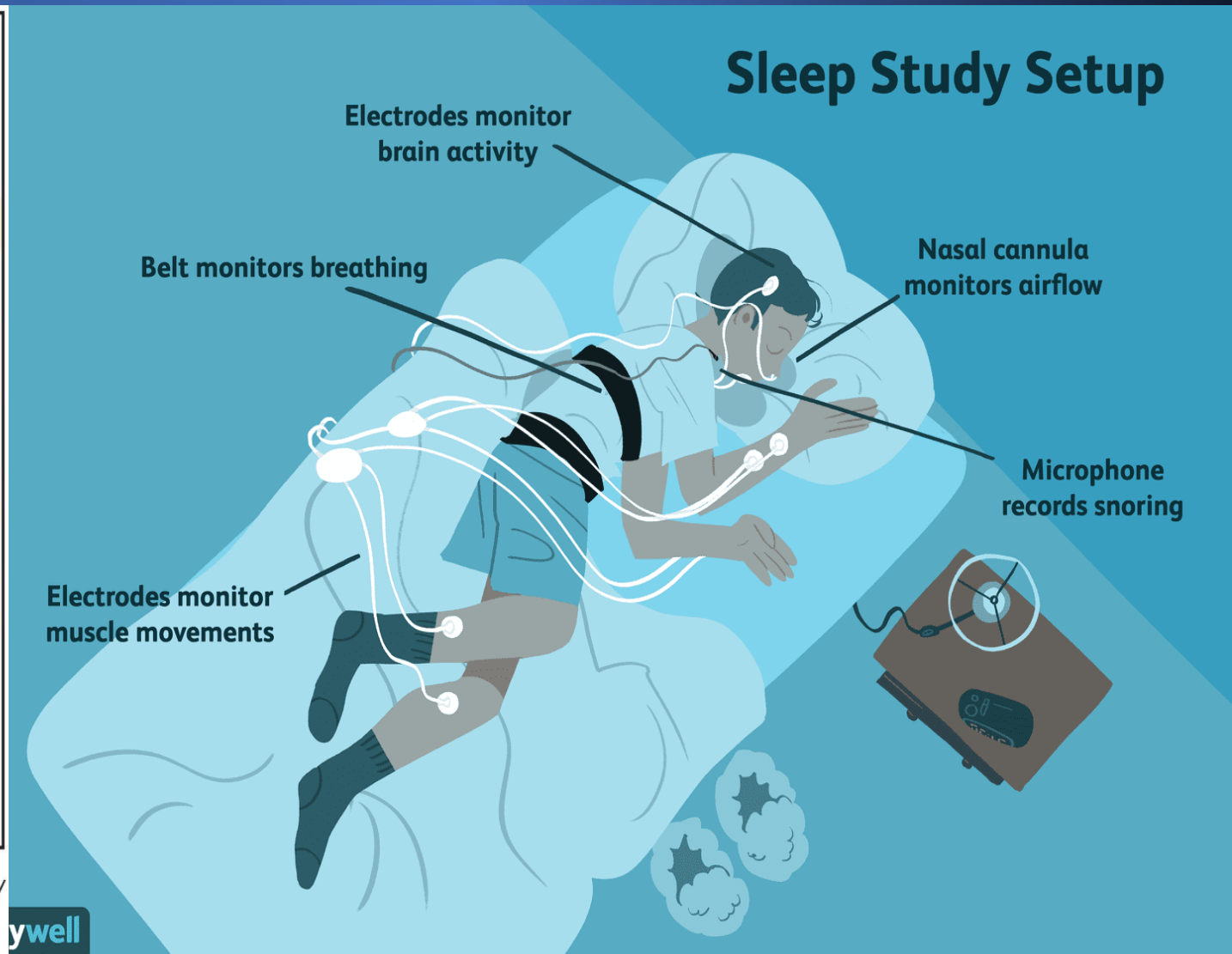
Please answer the following questions by checking "yes" or "no" for each one.

	Yes	No
Snoring (Do you snore loudly?)	❑	❑
Tiredness (Do you often feel tired, fatigued, or sleepy during the daytime?)	❑	❑
Observed Apnea (Has anyone observed that you stop breathing, or choke or gasp during your sleep?)	❑	❑
High Blood **P**ressure (Do you have or are you being treated for high blood pressure?)	❑	❑
BMI (Is your body mass index more than 35 kg per m^2?)	❑	❑
Age (Are you older than 50 years?)	❑	❑
Neck Circumference (Is your neck circumference greater than 40 cm [15.75 inches]?)	❑	❑
Gender (Are you male?)	❑	❑

Score 1 point for each positive response.

Scoring interpretation: 0 to 2 = low risk, 3 or 4 = intermediate risk, ≥5 = high risk.

Source: University Health Network, Toronto, Ontario, Canada (www.stopbang.ca/osa/screening/php). Used with permission from Sauk Prairie Healthcare.



Obstructive Sleep Apnea (OSA)

Management:

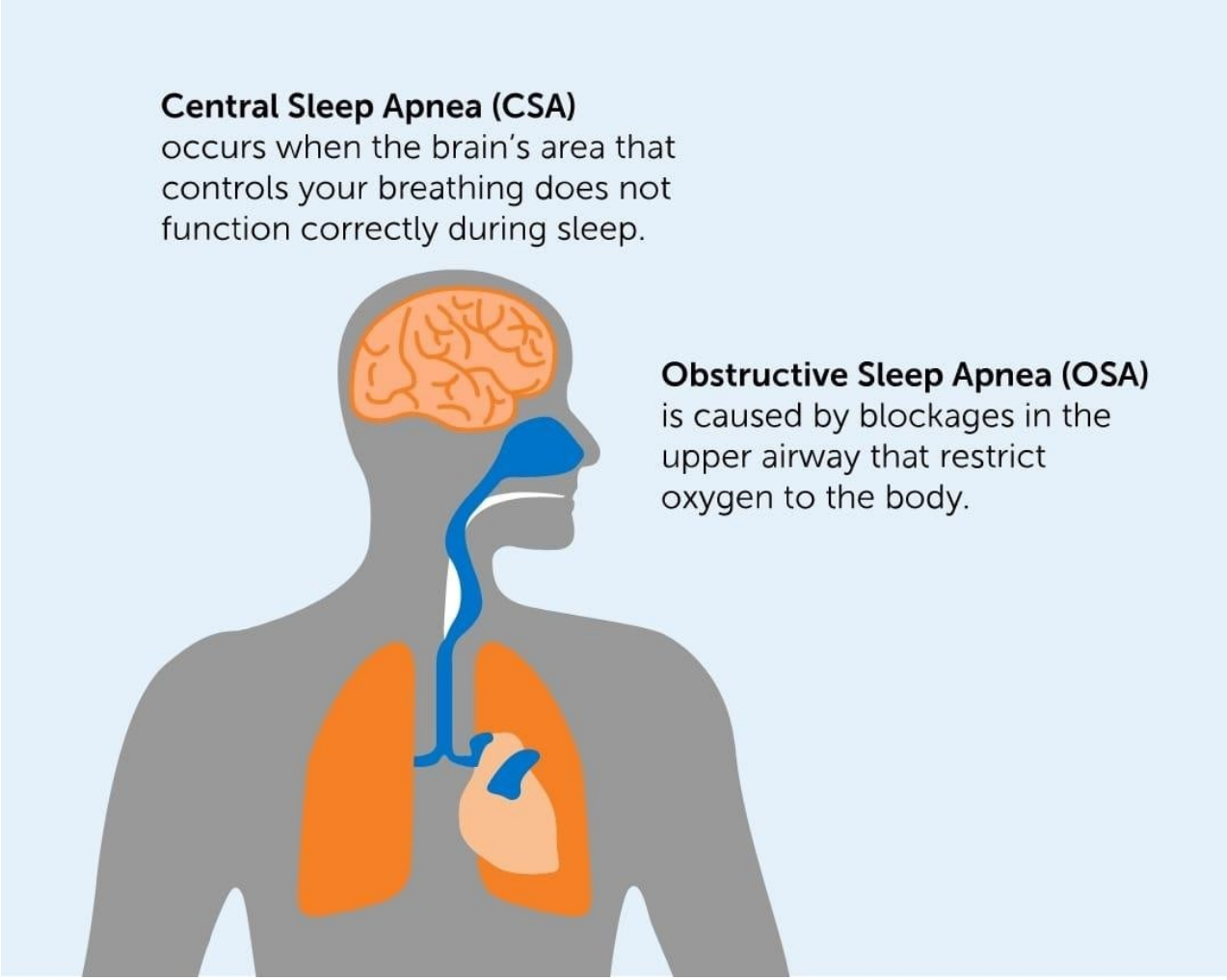
- weight loss and exercise
- Stop smoking
- avoiding alcohol and sedatives
- Oral appliance maintain mandibular advancement and prevent airway collapse
- sleeping in a nonsupine position
- **Continuous positive airway pressure (CPAP) first line**
- Uvulopalatopharyngoplasty (selective patients)
- Tracheostomy (last resort in cases of severe disease)



Central Sleep Apnea (CSA)

Definition:

sleep-related breathing disorder in which the drive to breathe periodically decreases or ceases due to impaired function of the respiratory center.



Central Sleep Apnea (CSA)

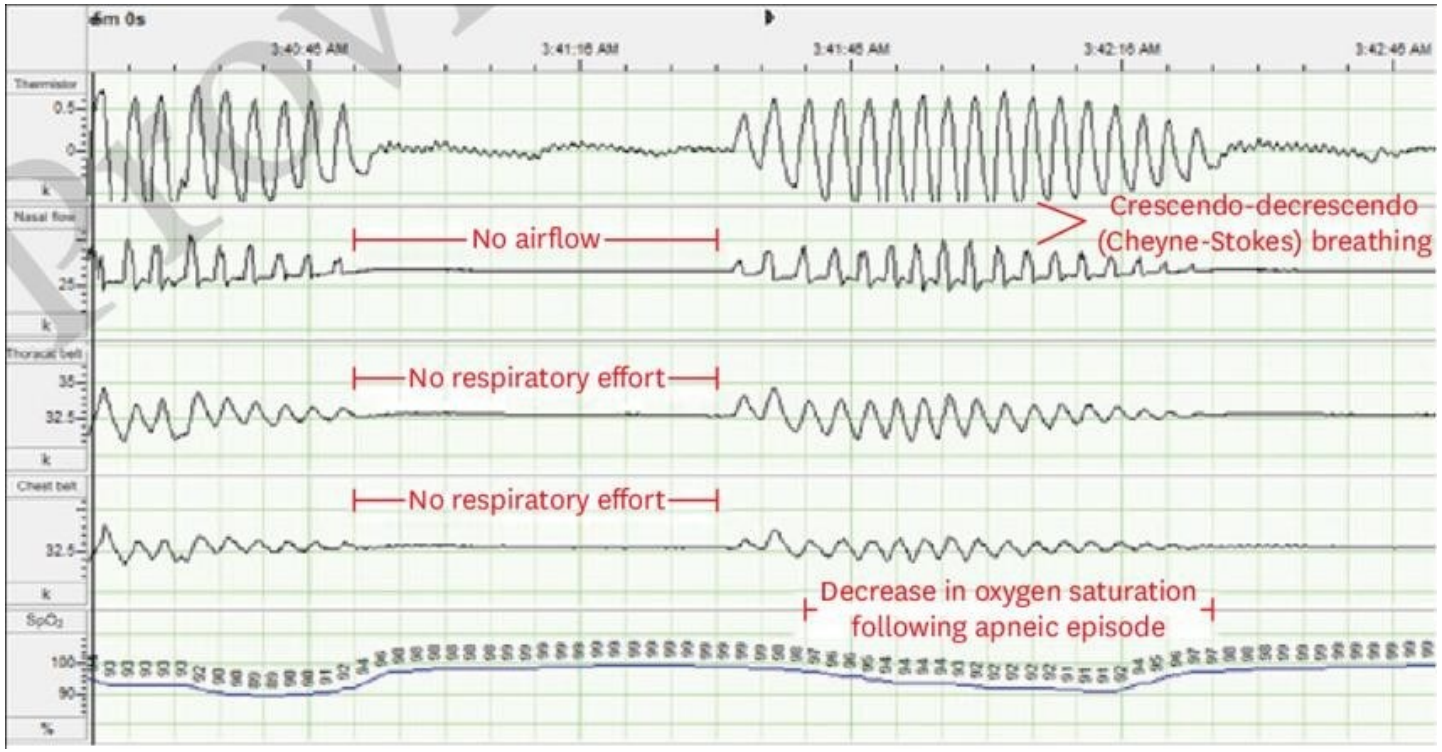
Central Sleep Apnoea

	Hypocapnia	**Hypercapnia**	
Mechanisms	Hyperventilation	Hypoventilation -	
Category/ prevalence	Primary CSA - *prevalence low*	Obesity	
	Cheyne Stokes Breathing Pattern - *prevalence high*	Neuromuscular Opiates Neurodegenerative diseases *prevalence unknown*	→ Stroke Multisystem atrophy Parkinson Alzheimer
Treatment	CPAP Acetazolamide O_2, CO_2 Theophylline	NIPPV with backup rate	

Central Sleep Apnea (CSA)

Signs & Symptoms:

- Similar to OSA
- Cheyne-Stokes breathing pattern

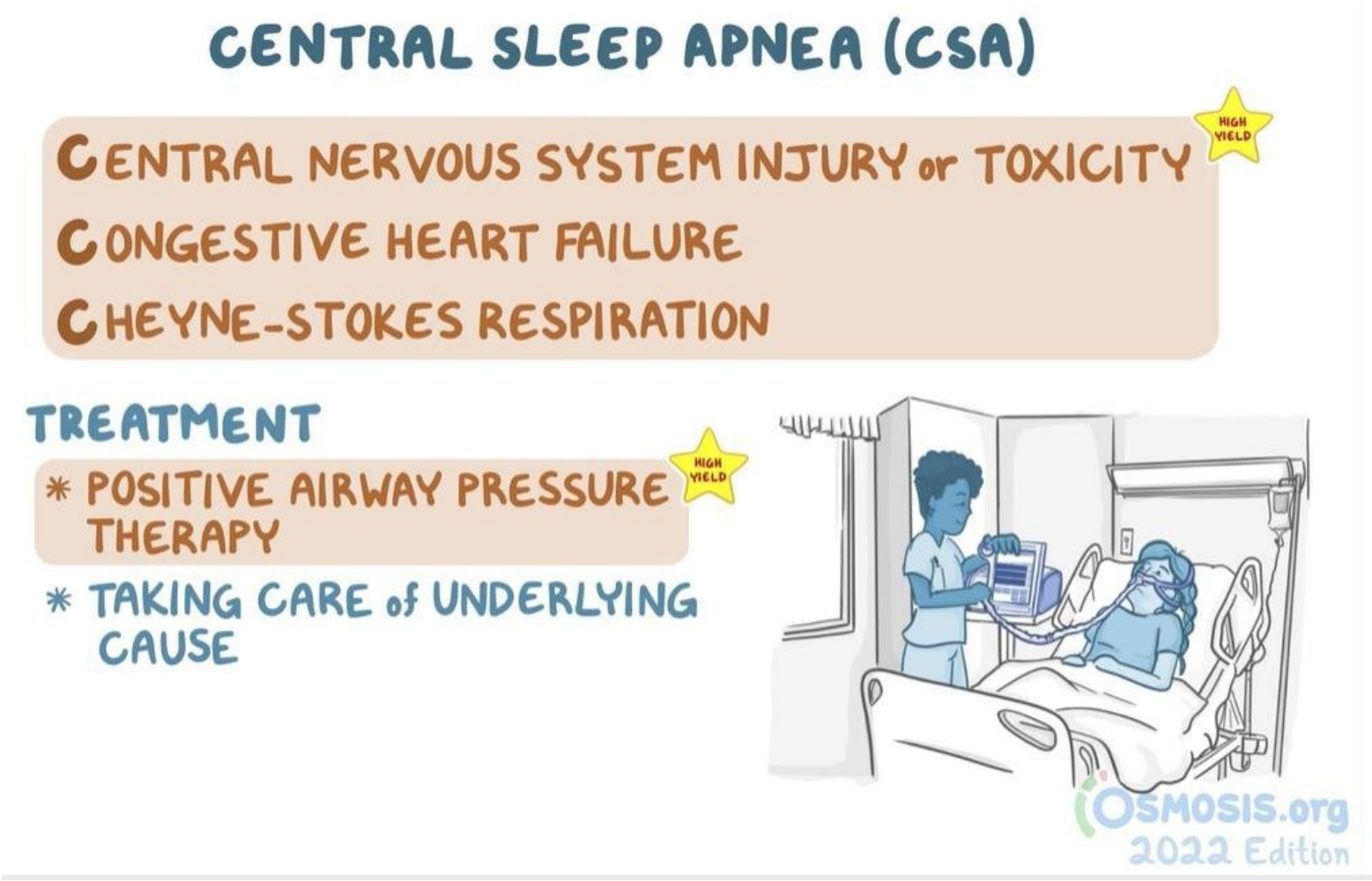


Central Sleep Apnea (CSA)

- **Diagnostics**
- Clinical history of underlying conditions (e.g., HF, stroke)
- Polysomnography, ideally with esophageal pressure manometry (gold standard)

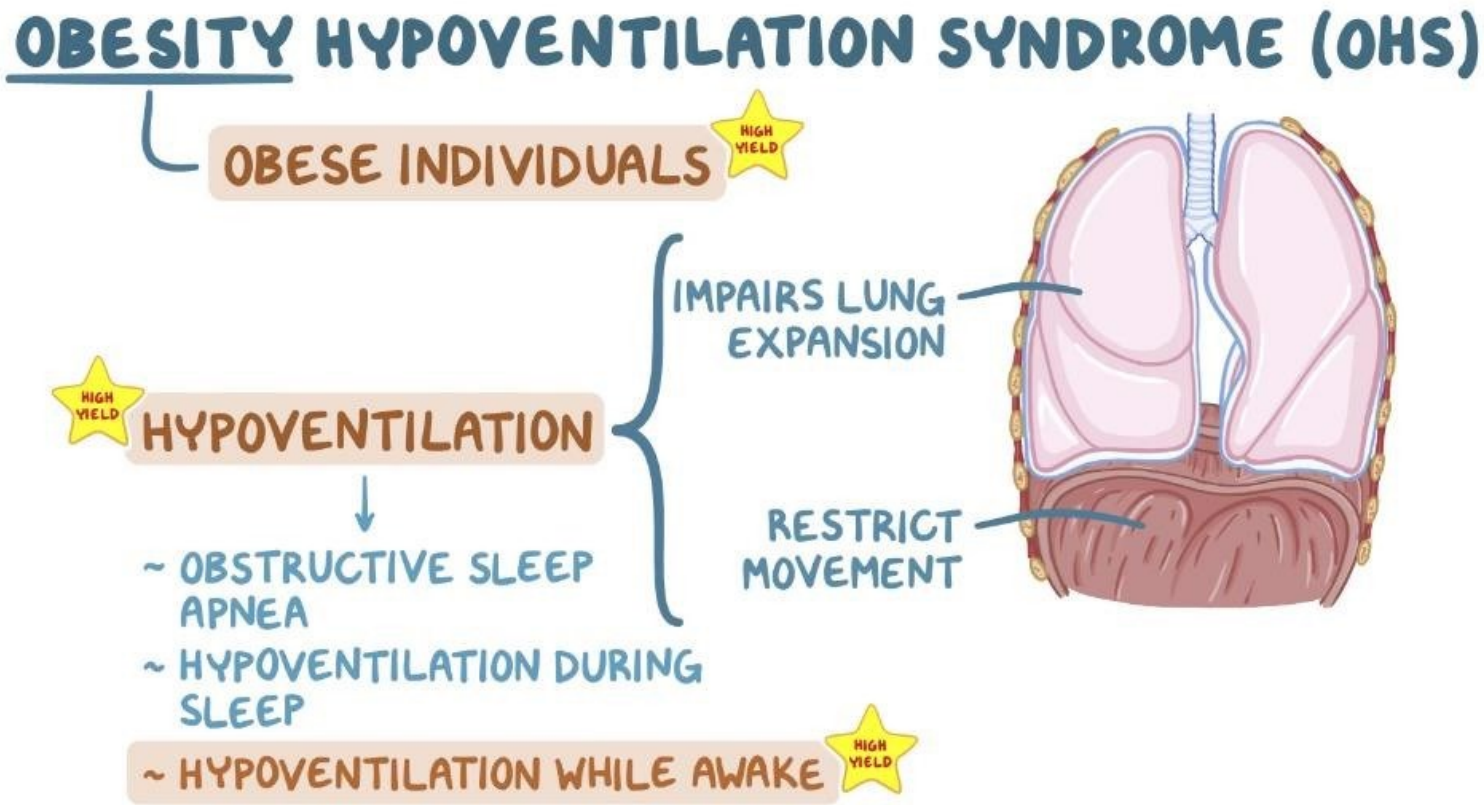
Central Sleep Apnea (CSA)

Management:



Obesity Hypoventilation Syndrome (OHS)
Pickwickian Syndrome

Overview:



Obesity Hypoventilation Syndrome (OHS)
Pickwickian Syndrome

What is Obesity Hypoventilation Syndrome?

Obesity Hypoventilation Syndrome (OHS) is a medical disorder in which a person has trouble breathing owing to excessive weight.



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Obesity Hypoventilation Syndrome (OHS)
Pickwickian Syndrome

Definition: a type of sleep-related hypoventilation disorder defined by a BMI of ≥ 30 kg/m2, diurnal hypercapnia, and disordered breathing during sleep

•**Pathophysiology**:

Multiple mechanisms caused by obesity contributing to the development of OHS.

•Structural component: reduction in inspiratory muscle strength and restriction of respiratory excursions → alveolar hypoventilation

•Central component: increased work of breathing → increased respiratory drive → inability to maintain during REM sleep → hypoventilation during sleep → depression of central respiratory centers → diurnal hypercapnia

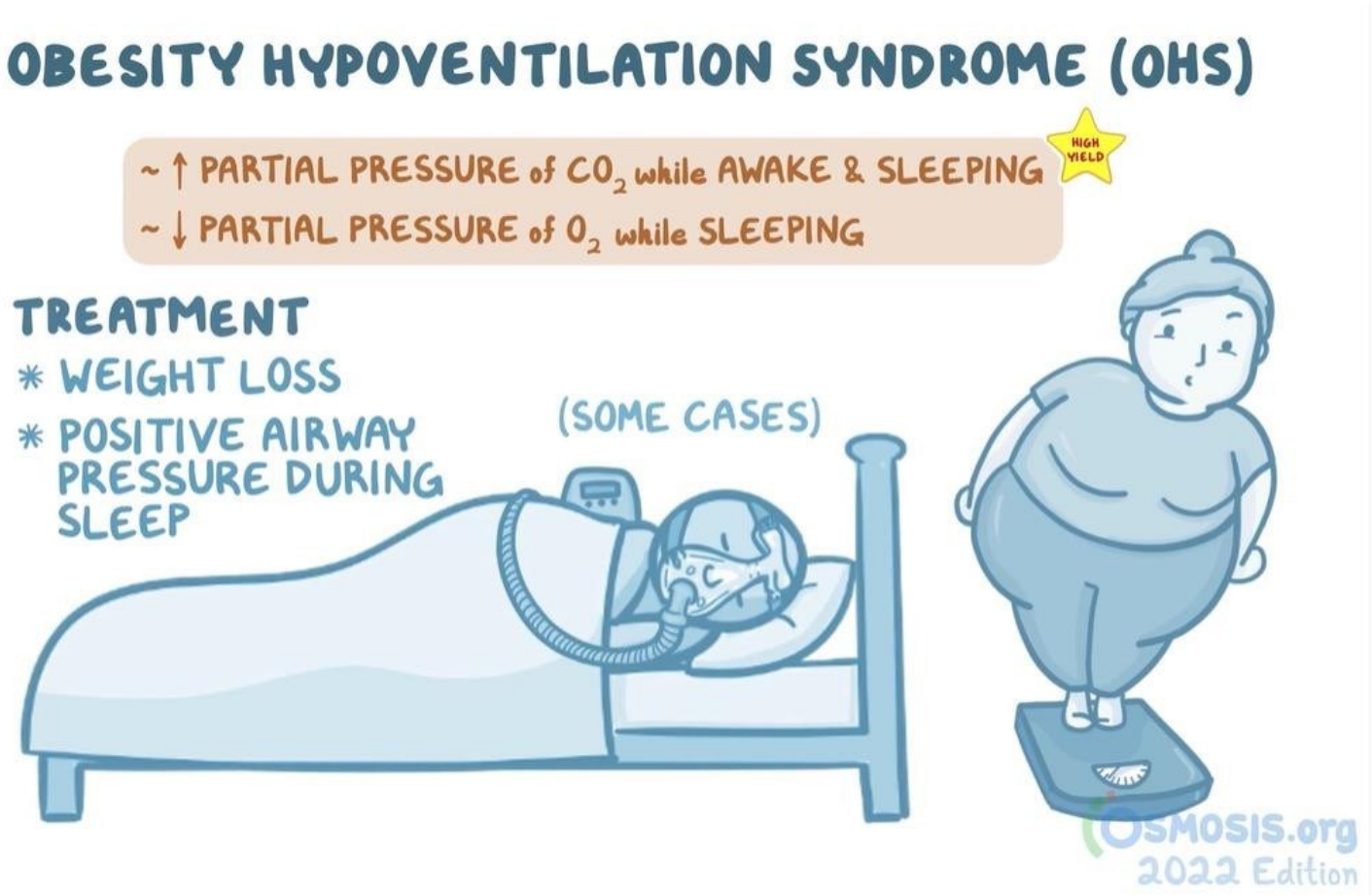
•Obstructive component: ~ 90% of OHS patients have concurrent OSA

Clinical features

- Similar to OSA, including headaches and severe sleepiness
- Possibly dyspnea during the day
- Often diagnosed in acute state of respiratory failure

Obesity Hypoventilation Syndrome (OHS)

Management:



To Summarize

SLEEP APNEA

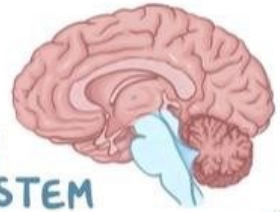
* INDIVIDUAL INTERMITTENTLY STOPS BREATHING for at least 10 SECONDS DURING SLEEP
  - SNORING
  - DISRUPTIVE SLEEP
  - DAYTIME SLEEPINESS
  - NOCTURNAL HYPOXIA → ~ HYPERTENSION, ~ ARRHYTHMIAS, ~ SUDDEN DEATH





CENTRAL

* IMBALANCE of RESPIRATORY CENTER in BRAIN DURING SLEEP
* CENTRAL NERVOUS SYSTEM TOXICITY or INJURY
* CONGESTIVE HEART FAILURE
* CHEYNE-STOKES RESPIRATION



OBSTRUCTIVE

* OBESE INDIVIDUALS
* AIRWAY OBSTRUCTION
* LOUD SNORING



Case 1

A 56-year-old man comes to the physician for a 5-month history of progressive bilateral ankle swelling and shortness of breath on exertion. He can no longer walk up the stairs to his bedroom without taking a break. He also constantly feels tired during the day. His wife reports that he snores at night and that he sometimes chokes in his sleep. The patient has smoked 1 pack of cigarettes daily for 25 years. He has a history of hypertension treated with enalapril. His pulse is 72/min, respirations are 16/min, and blood pressure is 145/95 mm Hg. There is jugular venous distention. The lungs are clear to auscultation bilaterally. The extremities are warm and well perfused. There is 2+ lower extremity edema bilaterally. ECG shows right axis deviation. Which of the following is the most likely cause of this patient's condition?

a. Left ventricular hypertrophy
b. Ischemic heart disease
c. Chronic hypoxia
d. Chronic kidney damage
e. Pulmonary neoplasia
f. Alveolar destruction

Questions?



References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board 4th edition
- Osmosis

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Thank You